

BUSINESS IDEA RISK SIMULATOR
TEST REPORT

Version 1.0

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1. Introduction

1.1. Definitions of Abbreviations

Table 1.1

Abbreviation	Explanation
BIRS	Business Idea Risk Simulator
HTML	HyperText Markup Language — used to structure the web page
CSS	Cascading Style Sheets — used for visual styling of the interface
JS	JavaScript — used for all interactive logic and calculations
Risk Index	Aggregated numerical score (0–100) representing the overall business idea risk
Low Risk	Risk Index value between 0 and 30
Medium Risk	Risk Index value between 31 and 70
High Risk	Risk Index value between 71 and 100
Weighted Scoring	Risk calculation method based on slider parameter values mapped to a 0–100 scale

1.2. Document Purpose

This document describes the test report for the study project **Business Idea Risk Simulator** (hereinafter “BIRS”).

The document is intended for the parties involved in testing and evaluation of the software developed within the study project:

- Lecturers of the customer, who are responsible for accepting and evaluating the project deliverables.
- The developer, who is responsible for the implementation of the system.

1.3. Related Documents

This test report has been developed based on the following documents:

1. Software Requirements Specification — BIRS
2. System Design Description — BIRS

3. BIRS Prototype (index.html)

1.4. General Requirements for Test Execution

In order to test the operation of the system, the tester needs:

- A modern web browser (Google Chrome, Mozilla Firefox, or Microsoft Edge)
- The BIRS prototype file (index.html) available locally on the machine
- A running local HTTP server (e.g. Python HTTP server on port 8000)
- No internet connection required — the system runs entirely in the browser
- No user account or login credentials are required

2. Functions That Need Testing

Table 2.1

No.	Description	Requirement Reference
1	Idea Submission Form	BIRS-UI-01
2	Input Validation	BIRS-UI-02
3	Logo Generation	BIRS-UI-03
4	Similar Business Lookup	BIRS-UI-04
5	Risk Calculation Engine	BIRS-RE-01, BIRS-RE-02, BIRS-RE-03
6	Risk Analysis Dashboard	BIRS-RP-01
7	Detailed Risk Report	BIRS-RP-02

3. Test Scenarios

3.1. Idea Submission Form

Open the BIRS prototype by navigating to <http://localhost:8000> in the browser.

3.1.1. Verifying the page loads correctly

Steps:

1. Open <http://localhost:8000> in the browser.

Expected Result:

The page loads and displays the title “Business Idea Risk Simulator” along with the General Information and Risk Parameters sections.

Test Result: Passed

3.1.2. Verifying all input fields are present

Steps:

1. Inspect the General Information section.

Expected Result:

The following fields are visible:

- Business idea name
- Description
- Industry dropdown
- “Show similar businesses” button

Test Result: Passed

3.1.3. Verifying the Industry dropdown options

Steps:

1. Click on the Industry dropdown.

Expected Result:

A list of 17 industry options is displayed, including Technology, Healthcare, Education, Finance, Tourism, and others.

Test Result: Passed

3.1.4. Verifying slider functionality

Steps:

1. Move each of the six sliders from minimum to maximum position.

Expected Result:

Each slider label updates dynamically to reflect the selected value (e.g. “Very small” → “Very large”).

Test Result: Passed

3.2. Input Validation**3.2.1. Submitting the form with all fields empty****Steps:**

1. Leave the Business idea name field empty.
2. Leave the Description field empty.
3. Click the “Analyze Risk →” button.

Expected Result:

A red error message “Business idea name is required.” is displayed below the name field. The form is not submitted.

Test Result: Passed

3.2.2. Submitting with a description shorter than 5 words**Steps:**

1. Enter a valid business idea name.
2. Enter “short text” in the Description field.
3. Click the “Analyze Risk →” button.

Expected Result:

A red error message “Please write a meaningful description (at least 5 words).” is displayed. The form is not submitted.

Test Result: Passed

3.2.3. Submitting with whitespace-only input in the name field**Steps:**

1. Enter several spaces in the Business idea name field.
2. Enter a valid description.

3. Click the “Analyze Risk →” button.

Expected Result:

Whitespace-only input is treated as empty. The error message for the name field is displayed.

Test Result: Passed

3.2.4. Submitting a valid form

Steps:

1. Enter a valid business idea name (e.g. “Didem Bakery”).
2. Enter a description with at least 5 words (e.g. “I want to open a bakery in Riga, Latvia”).
3. Click the “Analyze Risk →” button.

Expected Result:

No validation errors are shown. The risk analysis proceeds and results are displayed.

Test Result: Passed

3.3. Logo Generation

3.3.1. Logo appears when typing the business name

Steps:

1. Type “Didem Bakery” in the Business idea name field.

Expected Result:

A logo card appears above the name field, showing the initials “DB” on a colored square background. The business name and selected industry are displayed next to it.

Test Result: Passed

3.3.2. Logo initials update when name changes

Steps:

1. Change the Business idea name to “Travel Hub”.

Expected Result:

The logo initials update to “TH” automatically.

Test Result: Passed

3.3.3. Logo color changes with industry selection**Steps:**

1. Select “Tourism & Travel” from the Industry dropdown.
2. Change the selection to “Healthcare”.

Expected Result:

The logo background gradient color changes to reflect the selected industry each time.

Test Result: Passed

3.4. Similar Business Lookup**3.4.1. Searching with a European location in the description****Steps:**

1. Enter “I want to open a bakery in Riga, Latvia” in the Description field.
2. Click the “Show similar businesses →” button.

Expected Result:

The similar businesses box appears showing the label “EU European examples based on your description” and three European businesses related to food, each with:

- Name
- Country
- Description
- Clickable website link

Test Result: Passed

3.4.2. Searching with an American location in the description**Steps:**

1. Enter “I want to launch a food delivery app in New York, USA” in the Description field.
2. Click the “Show similar businesses →” button.

Expected Result:

The results show American businesses (e.g. DoorDash, Grubhub) with a “🌐 American examples” label.

Test Result: Passed

3.4.3. Searching with no matching keywords

Steps:

1. Enter “aaa bbb ccc ddd eee” in the Description field.
2. Click the “Show similar businesses →” button.

Expected Result:

A message is displayed:

“No similar businesses found. Try describing your idea with more specific keywords.”

Test Result: Passed

3.4.4. Verifying that matching is based on description only

Steps:

1. Type “DoorDash” in the Business idea name field.
2. Enter “I want to open a software company in Berlin, Germany” in the Description field.
3. Click the “Show similar businesses →” button.

Expected Result:

Results show technology companies from Europe, not food delivery companies. The business name field has no effect on the results.

Test Result: Passed

3.4.5. Verifying website links open correctly

Steps:

1. Perform a valid search that returns results.
2. Click one of the website links.

Expected Result:

The link opens in a new browser tab and leads to the correct company website.

Test Result: Passed

3.5. Risk Calculation Engine

3.5.1. Low risk scenario — all sliders at minimum

Steps:

1. Set all six sliders to their minimum position (value 1).
2. Fill in required fields and click “Analyze Risk →”.

Expected Result:

Risk Index = 0. Classification displayed as “Low Risk” with a green badge.

Test Result: Passed

3.5.2. High risk scenario — all sliders at maximum

Steps:

1. Set all six sliders to their maximum position (value 5).
2. Fill in required fields and click “Analyze Risk →”.

Expected Result:

Risk Index = 100. Classification displayed as “High Risk” with a red badge.

Test Result: Passed

3.5.3. Medium risk scenario — mixed slider values

Steps:

1. Set:
 - Market size = 3
 - Competition = 3
 - Budget = 3
 - Team experience = 3

- Technical complexity = 3
 - Audience = 3
2. Fill in required fields and click “Analyze Risk →”.

Expected Result:

Risk Index falls within the 31–70 range. Classification displayed as “Medium Risk” with an orange badge.

Test Result: Passed

3.5.4. Verifying three category scores are calculated

Steps:

1. Submit a valid form with any slider values.

Expected Result:

Three separate scores are displayed:

- Market Risk
- Resource Risk
- Adoption Risk

Each score is between 0 and 100.

Test Result: Passed

3.5.5. Verifying results update on resubmission

Steps:

1. Submit the form once and observe the results.
2. Change slider values and submit again.

Expected Result:

All scores and the Risk Index update correctly to reflect the new slider values.

Test Result: Passed

3.6. Risk Analysis Dashboard

3.6.1. Results section appears after submission

Steps:

1. Submit a valid form.

Expected Result:

The results section becomes visible below the form and the page scrolls smoothly to it.

Test Result: Passed

3.6.2. Overall Risk Index is displayed

Steps:

1. Submit a valid form and observe the results card.

Expected Result:

A large numerical Risk Index (0–100) is shown along with the risk level badge (Low / Medium / High).

Test Result: Passed

3.6.3. Category score cards are displayed

Steps:

1. Observe the three score cards below the Risk Index box.

Expected Result:

Market Risk, Resource Risk, and Adoption Risk scores are each shown with:

- A numerical value
- A progress bar proportional to the score

Test Result: Passed

3.6.4. Page subtitle updates after submission

Steps:

1. Submit a form with the business idea name “Didem Bakery”.

Expected Result:

The subtitle changes from:

“Fill in your business idea details and analyze its risk”

to:

“Risk Analysis Complete — Didem Bakery”

Test Result: Passed

3.7. Detailed Risk Report

3.7.1. Report section appears after submission

Steps:

1. Submit a valid form and scroll to the bottom of the results.

Expected Result:

A “Detailed Risk Report” section is visible with three report items:

- Market Risk
- Resource Risk
- Adoption Risk

Test Result: Passed

3.7.2. Low risk report shows green styling

Steps:

1. Set all sliders to minimum and submit.

Expected Result:

All three report items show a green left border. The text describes favorable conditions and the recommendation is growth-oriented.

Test Result: Passed

3.7.3. High risk report shows red styling

Steps:

1. Set all sliders to maximum and submit.

Expected Result:

All three report items show a red left border. The text warns about critical risk factors and the recommendation is defensive.

Test Result: Passed

3.7.4. Medium risk report shows orange styling**Steps:**

1. Set sliders to middle values and submit.

Expected Result:

All three report items show an orange left border with moderate analysis text.

Test Result: Passed

3.7.5. Each report item shows a recommendation**Steps:**

1. Submit any valid form and inspect the report items.

Expected Result:

Each report item displays:

- An analysis paragraph
- A recommendation line starting with “→”

Test Result: Passed